# Summary of findings

Cleaned, merged dataset: **25,300** students (after dropping nulls and removing 1,427 duplicate rows from part_2a).
`click_events` present for **23,984** students (1,331 missing in the original part_2b before dropping).

## High-level descriptive results

- **Score distribution**
  - Mean score ≈ **73.4**, SD ≈ **15.5** (min 0, max 100).
  - **1,876** students scored **< 50**.
  - **520** outliers detected by the 1.5×IQR rule (kept intentionally).
- **Final outcomes**
  - `Fail`: **5,654** students.
  - `Withdrawn`: **4,848** students.

## Group differences

- **By highest_education (mean score, high → low)**
  - Post Graduate Qualification: **82.16**
  - HE Qualification: **75.75**
  - A Level or Equivalent: **74.09**
  - Lower Than A Level: **71.51**
  - No Formal quals: **66.49**
- → Clear monotonic relationship: higher formal education → higher mean score.
- **By age_band (mean score)**

  - `0-35`: **72.61**
  - `35-55`: **75.02**
  - `55<=`: **77.27**
- → Older age groups tend to have higher mean scores.

## Statistical tests & assumptions

- **Normality** (Anderson–Darling): test statistic ≈ **361.01**; strongly rejects normality.
  → score distribution is **not normal**.
- **Kruskal–Wallis** (non-parametric test across age bands):
  H ≈ **196.26**, p ≈ $\mathbf{2.41 \times 10^{-43}}$ → **reject null**.
  → There is **strong evidence** that score distributions differ between at least one pair of age groups.

## Important caveats & data issues

- You dropped rows with any nulls: this removed some `click_events` and a few `score` rows (19 score nulls originally). Consider impact on representativeness.
- `click_events` had **1,331** missing values originally — investigate whether missingness is random.
- `id_student` duplicates in part_2a were removed as full-row duplicates; if those duplicates represented legitimate repeated measurements, that would change results.

## Practical interpretation

- Education background and age are associated with performance: more educated and older students score higher on average.
- A non-trivial proportion of students (≈7.4% of merged dataset) score <50 and many either fail or withdraw — possible targets for intervention.
- Because the score distribution is non-normal, non-parametric tests and robust summaries are appropriate.